

Task 7HD

The goal for this HD task is to reproduce the main results of a research paper [“Comparison of Machine Learning Algorithms for the Power Consumption Prediction: Case Study of Tetouan City”](#) and propose your own approach to do further improvement. You are suggested to read the paper carefully and get familiar with the [dataset](#) before doing the following tasks.

Execute your code into a jupyter notebook (.ipynb file) and keep the output, write a report (.pdf file) to answer the following questions, and submit both files - your code and your report.

1. Read the article and reproduce the results presented in Table II and Table IV using Python modules and packages (including your own script or customised codes). Write a report summarising the dataset, used ML methods, experiment protocol and results including variations, if any. During reproducing the results:

- i) use the same set of features used by the authors.
- ii) use the same classifier with exact parameter values.
- iii) use the same training/test splitting approach as used by the authors.
- iv) use the same pre/post processing, if any, used by the authors.
- v) report the same performance metric (RMSE, and MAE) as shown in Table II and Table IV.

2. Design and develop your own ML solution for this problem. The proposed solution should be different from all approaches mentioned in the provided article. This does not mean that you must have to choose a new ML algorithm.

You can develop a novel solution by changing the feature selection approach or parameter optimisations process of used ML methods or using different ML methods or adding regularization or different combinations of them.

This means, **the proposed system should be substantially different from the methods presented in the article but not limited to only change of ML methods.**

Compare the RMSE/MAE result with reported methods in the article. Write in your report summarising your solution design and outcomes. The report should include:

- i) Motivation behind the proposed solution.
- ii) How the proposed solution is different from existing ones.
- iii) Detail description of the model including all parameters so that any reader can implement your model.
- iv) Description of experimental protocol.
- v) Evaluation metrics.
- vi) Present results using tables and graphs.
- vii) Compare and discuss results with respect to existing literature.
- viii) Appropriate references ([IEEE numbered](#)).

Assessment feedback

The results with comments will be released within 5 business days from the due date.

Referencing

You must correctly use the IEEE method in this assessment. See the [referencing guide](#).

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